dewlock

• SUI DEFI · INTENT FIREWALL

Every transaction, sealed before you sign.

State a goal in plain language. A keyless agent compiles it into one unsigned PTB; a deterministic, fail-closed Guardian re-derives the math and dry-runs that exact bundle — and you sign the literal artifact you reviewed.

0	**11**	**620**
USER KEYS SERVER-SIDE	FAIL-CLOSED GATES	TESTS GREEN

Sui Mainnet · Live
dewlock.vercel.app

github.com/tienlenan/dewlock

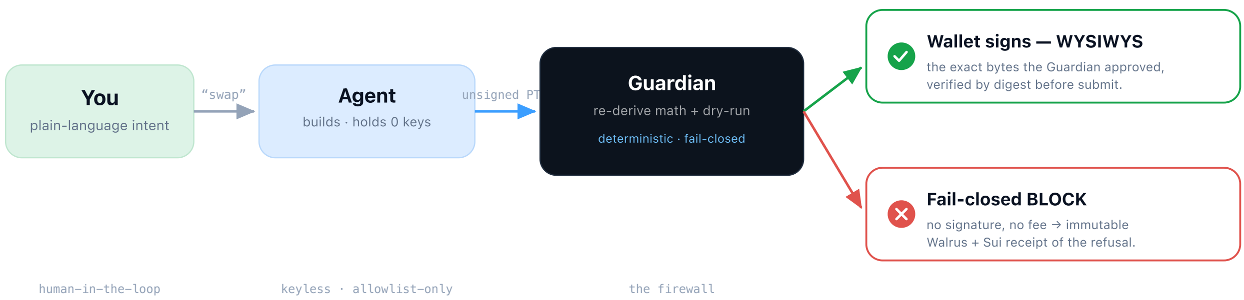
THE PROBLEM & THE MODEL

Acting on Sui DeFi today means trusting a black box.

Liquidity is scattered across Cetus, DeepBook, SuiNS. The natural-language layer everyone wants usually means handing an agent your keys, or signing bytes you never read. Dewlock inverts it: the agent **builds**, the Guardian **verifies**, and **you sign** the exact artifact — never a key on the server.

Intent → Sign — the firewall pipeline

flow · 01



Your device — keys never leave · Keyless agent — builds only · Guardian — deterministic gates

Approved → sign · Blocked → receipt

HONEST BY DESIGN

Custody	0 user keys server-side
Identity	coin TYPE, not symbol
Valuation	trusted price · not pool
Limits	$5 / tx · $20 / day

THE HEADLINE

"An agent that holds **zero user keys**, remembers your rules on decentralized storage, and is **provably blocked** from the wrong thing before you sign."

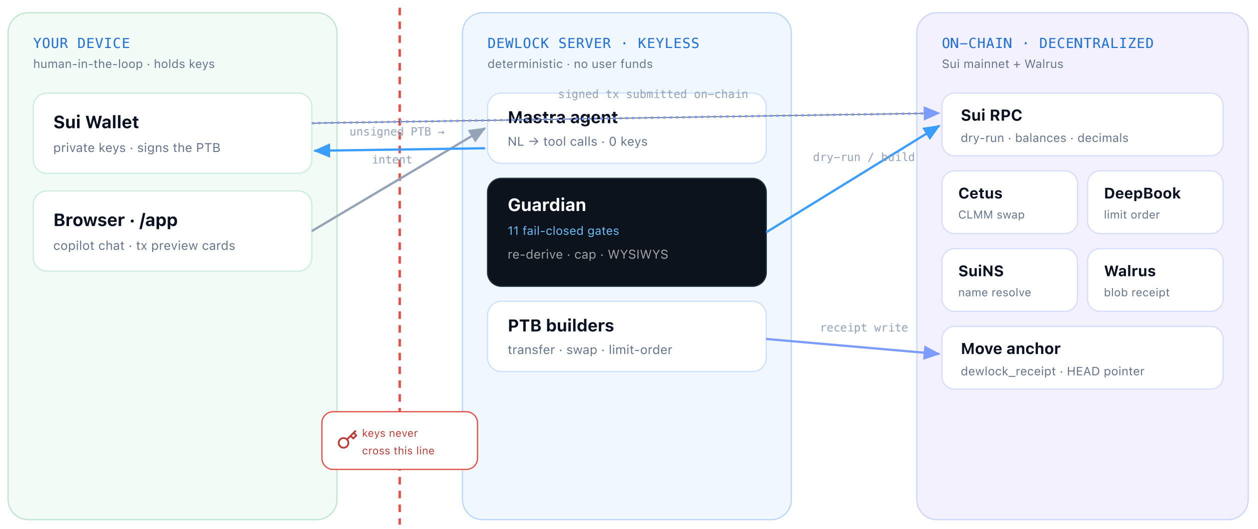
• SYSTEM ARCHITECTURE

Three trust zones, one boundary that keys never cross.

The agent and Guardian run server-side but stay **keyless** — they only ever produce an *unsigned* bundle. Signing happens in your wallet; settlement, valuation and receipts happen on-chain.

Component & data-flow map

architecture · 02



Device — holds keys   Server — keyless / deterministic   On-chain — Sui + Walrus

Trust boundary (no keys cross)   Unsigned build / dry-run

● WHAT MAKES IT DIFFERENT

Three things, each impossible to fake.

Not “chat that swaps.” Real on-chain order work plus adversarial-grade correctness — and a Guardian every action must pass through.

THE MOAT

Fail-closed Guardian

Deterministic gates — never an LLM. Caps, trusted-price valuation, WYSIWYS digest binding, min-out re-derived on-chain, price-impact, allowlist, lookalike, coin-type.

`defends the math-failure exploit class`

THE NOVELTY

DeepBook limit-order

“Wait at my price, don’t market-buy.” A POST_ONLY `placeLimitOrder` on the same spine — self-match, expiry & BalanceManager gates. Impossible on an AMM.

`true limit orders no AMM can offer`

THE CLIMAX

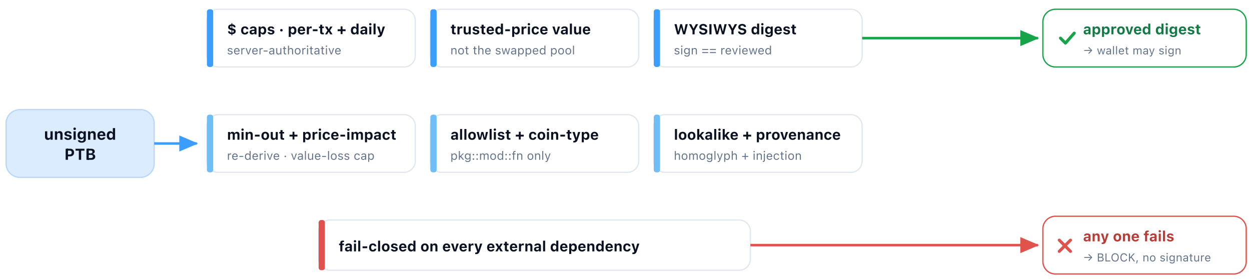
The provable BLOCK

A deliberate fail-closed block — lookalike `888-l.sui` + broken min-out — refused before signature, then an immutable Walrus + Sui receipt.

`proof a BLOCK happened, not just a tx`

The Guardian gate-stack — every PTB runs the gauntlet

flow · 03



Value & integrity gates · Identity & correctness gates · Dependency fail-closed · All pass → sign · Any fail → block

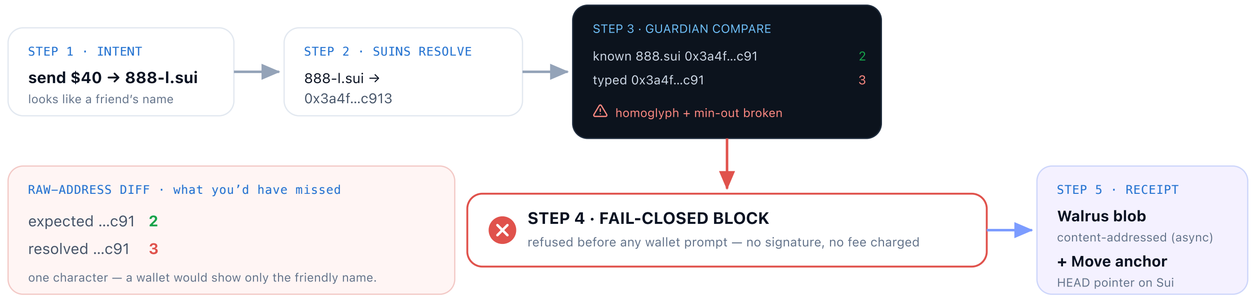
THE HERO BEAT

The BLOCK, drawn step by step.

A lookalike SuiNS name (`888-l.sui`) and a broken min-out walk into the Guardian. It refuses **before any wallet prompt**, shows the raw-address diff, then writes a tamper-evident receipt — proof a block happened.

Fail-closed block → immutable receipt

sequence · 04



Guardian compare Lookalike mismatch On-chain receipt Fail-closed path

UNDER THE HOOD & WHERE IT STANDS

Built on Sui & Walrus primitives that couldn't exist elsewhere.

A single Next.js app + a keyless Mastra agent + Guardian-as-code + your wallet. Persistent memory and tamper-evident receipts on Walrus, anchored on Sui.

SHIPPED & VERIFIED

Guardian spine	11 gates · reviewed
Swap · lend · transfer · track	Cetus · Aftermath · NAVI
DeepBook limit-order	POST_ONLY + gates
BLOCK + receipt	Move pkg + async blob
Seal-encrypted chat	owner-only decrypt
Tests · CI · deploy	620 green · GH · Vercel

WHY SUI-NATIVE

PTB composition
atomic multi-leg · one signature

Walrus + memwal
tamper-evident receipts + memory

SuiNS · Seal
named sends · owner-only chat

STACK

Next.js 16 · React 19 · TypeScript · Tailwind v4 · @mysten/sui · dapp-kit · @mysten/deepbook-v3 · Cetus · Aftermath · NAVI · Suilend · @mysten/suins · Wormhole · Walrus · memwal · Seal · Mastra · AI Gateway · Claude · Gemini · Move

mainnet · dewlock_receipt 0x8c3b…2361 · dewlock_seal 0x77aa…0709 · live at dewlock.vercel.app

DEMO ARC · ~4 MIN, ONE E2E FLOW

Connect & track → atomic 2-leg PTB (sign once) → DeepBook "wait at my price" → **the BLOCK** (lookalike + broken min-out, refused before signature) → tap → immutable Walrus + Sui receipt → "send $40" → the agent recalls your day-1 $5 cap and freezes it. *Zero user keys, real on-chain order work, provably blocked before you sign.*